Lung-protective ventilation in the operating theatre with Atlan, even for the smallest patients

Experience report - The Paediatric and Youth Hospital Auf der Bult, Hannover

Dräger

Technology for Life

The team at The Paediatric and Youth Hospital Auf der Bult in Hannover places a special focus on protective ventilation in the operating theatre. With the Atlan anaesthesia machine from Dräger, this can be carried out safely even in the smallest patients. As a result, the hospital no longer uses a neonatal intensive care ventilator in the operating theatre since acquiring the Atlan.

From 500 grams to over 150 kilograms in body weight, from the smallest premature babies to tall adolescents with severe obesity approaching adulthood – a wide variety of patients undergo surgery at the Paediatric and Youth Hospital Auf der Bult in Hannover. The hospital has been using Atlan anaesthesia machines for all surgeries since 2022. These machines were purchased following a thorough comparison and testing phase.

"We have a very wide range of patients," confirms Prof. Dr. Christoph Bernhard Eich, Head of the Department of Anaesthesia, Paediatric Intensive Care and Emergency Medicine. The hospital opened in 1983 and began treating this full spectrum of patients in 2011. In addition to the broad spectrum of ages, heights and weights, a high proportion of seriously ill or severely injured children are also among the patients. Professor Eich summarises the situation by stating that "all of this poses significant challenges for the medical devices we use."

Prof. Dr. Christoph Bernhard Eich (Chief Physician)

Patient and operational safety

"Before purchasing the Dräger Atlan anaesthesia machines, we also tested products from other manufacturers," summarises Michael Brackhahn. The specialist in anaesthesiology and Master of Health Business Administration (MHBA) is a senior physician and medical equipment officer in the Department of Anaesthesia, Paediatric Intensive Care and Emergency Medicine at the Paediatric and Youth Hospital Auf der Bult.

He describes the decision-making process as follows: "Initially, patient safety and our own occupational safety were the pivotal factors in our decision to purchase Dräger anaesthesia machines. The team then collectively chose the more compact anaesthesia machine, which features the latest generation of piston ventilators, capable of meeting all requirements—even for the smallest patients."

The Atlan anaesthesia machine was designed to administer very small tidal volumes and perform low-flow anaesthesia safely. Anaesthesiologist Michael Brackhahn describes the Volume-controlled AutoFlow (VC-AF) ventilation mode, predominantly used in Hannover, as "the best form of ventilation available for our specialty."

The Paediatric and Youth Hospital Auf der Bult offers a capacity of over 250 beds, spread across numerous wards and specialised departments. With a legacy in paediatric medicine that spans more than 150 years, the hospital is part of the non-profit Hannoversche Kinderheilanstalt foundation, continuing to provide essential care to sick children and adolescents. Furthermore, it also serves as an academic teaching hospital for Hannover Medical School.

The hospital offers a wide range of specialised medical services across various areas of paediatric medicine. The Department of Anaesthesia, Paediatric Intensive Care and Emergency Medicine provides perioperative anaesthetic care for approximately 5,000 cases each year – from premature and newborn babies to infants, children, and adolescents. Additionally, the department oversees the intensive care of around 500 patients each year.

Precise tidal volume delivery when it counts

The younger and smaller the patients, the more demanding ventilation becomes during surgery, says the senior consultant: "When ventilating premature and, in particular, extremely premature babies, we have to take their different physiology into account, especially the special lung physiology. We quickly encounter the limitations of the devices when attempting to reliably administer very low tidal volumes at relatively high respiratory rates," explains the anaesthesiologist. "That's why it's so important for us that the Atlan delivers and measures tidal volumes with high precision, even for our smallest patients, who require tidal volumes of just five millilitres."

Selecting the optimal ventilation technique and strategy necessitates meticulous consideration, especially in the perioperative care of neonates. Some clinics routinely use neonatal intensive care ventilators for the youngest patients in the operating theatre, offering advantages in terms of ventilation physiology.

However, this approach does not allow doctors the option to administer inhalation anaesthetics, which, according to Mr. Brackhahn, are considered a readily controllable and relatively gentle anaesthetic method for these patients. Consequently, the Atlan anaesthesia machine is predominantly used for surgeries involving even the smallest patients at the Paediatric and Youth Hospital Auf der Bult in Hannover. Exceptions are made for operations on particularly vulnerable neonates, which are conducted directly in the intensive care unit to avoid transport and repositioning.

Michael Brackhahn (Senior Physician)

”With the introduction of the Atlan at the Paediatric and Youth Hospital Auf der Bult in Hannover, there is no longer a need for a neonatal intensive care ventilator in the operating theatre," states Professor Eich.

Performance is tested and confirmed

After over two years of intensive use, Mr. Brackhahn has affirmed the performance of the Atlan. “This purchase was the right investment for us," says the senior physician. He specifically highlights the machine's ability to ventilate neonates which places high demands on the technology: “For very small patients, the dead space in the ventilation hoses is larger than the tidal volume."

In practice, the anaesthesia machine stands out for its exceptional CO_2 measurement performance, made possible by its stable and leak-tight system. Its role is especially critical during surgeries on premature babies, as blood gas analyses (BGA) can only be performed to a limited extent. The anaesthesiologist highlights several significant advantages of the Atlan, including its high trigger sensitivity, even with extremely weak spontaneous breathing, rapid gas exchange times, and the ability to operate with very low fresh-gas flows.

Efficiency in the operating theatre

It is not only the technical performance of ventilators that determines the success of medical care but also their simple and safe operation, even for very complex applications. “Its fantastic that the Atlan can ventilate even such small children so efficiently," says Stefanie Ohnesorge. The specialist nurse for anaesthesia and intensive care medicine has been working in the anaesthesia department since 2018 and uses the Atlan multiple times a day for both anaesthesia induction and operative procedures.

Stefanie Ohnesorge (Specialised health care and nursing assistant)

The wide range of patients is clearly reflected in daily clinical practice:"Depending on the day's surgical schedule, we might treat several very small children, or handle two major operations involving a neonate in the morning followed by a teenager weighing over 100 kilograms later in the day."

Mrs. Ohnesorge commends the device for its excellent ergonomics. She highlights several features of the device, including the the quick and easy preparation of the workstation for patients of varying ages and weights, its intuitive user interface, versatile attachment options for additional equipment and components, ample storage space, and the device's mobility—which is especially valuable given the limited space around the patient in the operating theatre.

Sabrina Obermeier, a paediatric nurse who has worked in anaesthesia at the Paediatric and Youth Hospital Auf der Bult since 2017, also points to key advantages of the Atlan in surgical practice. These include its intuitive operating principles, the clearly visible flow curve display – which changes colour during spontaneous breathing efforts – and its relatively compact design. "The Atlan is a compact anaesthesia machine. That's particularly important when working within the confined space of an operating theatre. You're simply glad for every centimetre of floor space you can save."

Progress for patients and medical personnel

With approximately 30 years of experience in anaesthesia, Prof. Eich views the Atlan technology as a significant advancement for both patients and medical personnel at the Paediatric and Youth Hospital: "I still remember the days when newborns, especially preterm infants, were ventilated with intensive care ventilators in the operating theatre. However, the surgical team wasn't as familiar with those devices as they were with anaesthesia machines—an obvious problem by today's standards. That's why I was relieved when pressure-controlled ventilators became available, allowing for effective adaptation to constantly changing patient sizes and weights."

The head physician emphasises the importance of a standardised operating principles across the equipment pool, considering it essential for establishing routine in clinical practice – a key factor in achieving successful treatment outcomes.

"It is imperative that we take such aspects into account, as demonstrated by the progress in neonatal medicine – not only at this hospital, but across the country. Complication rates have decreased significantly in recent years, even as the threshold for care continues to move forward. We are currently treating infants at 23+ weeks of gestational age. Prof. Eich affirms that medicine is clearly advancing in the right direction."

At the conclusion of the interview, Mr. Brackhahn smiles and gestures towards the cheerful image of Charlie Schlummer – an anaesthetist mascot who is part of a series of animal characters representing different medical roles. These mascots are prominently featured throughout the Paediatric and Youth Hospital Auf der Bult, fostering trust in doctors, nursing staff, and technology among the young patients.

Coffee mugs featuring endearing mascots

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